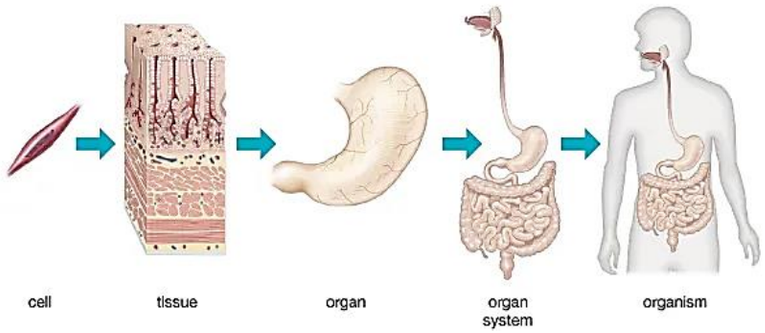
Knowledge Organiser
Year 7: Cells – Cell Structure

What are cells?
- Cells are the smallest units of life
- Without cells there would be no life
- Humans are **multicellular organisms**, they are made from many different cells
- Bacteria are **unicellular organisms,** they exist as individual cells

Levels of organisation within organisms:

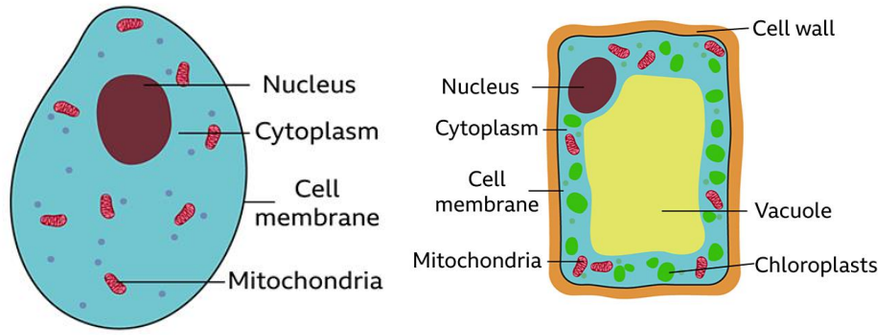


How are living things organised?
- All parts of a living organism fall into one of the **levels of organisation**
- Tissues are made from cells working together
- Organs are made from tissues working together
- Organ systems are made from many organs working together
- Organisms are a collection of all of these

Level	Examples
Cell	Sperm, Nerve cell, Red blood cell
Tissue	Bones, Muscle tissue, Blood
Organ	Lungs, Heart, Liver, Brain
Organ system	Digestive, Respiratory, Nervous
Organism	Dog, Human, Sunflower

What is inside an animal cell?
- Cells are made from parts called **organelles**
- Each type of cell will have a unique selection of organelles which perform different functions
- Typical animal cells have a: **nucleus**, **cell membrane**, **cytoplasm** and **mitochondria**

Labelled diagrams showing the organelles in an animal (left) and plant (right) cell



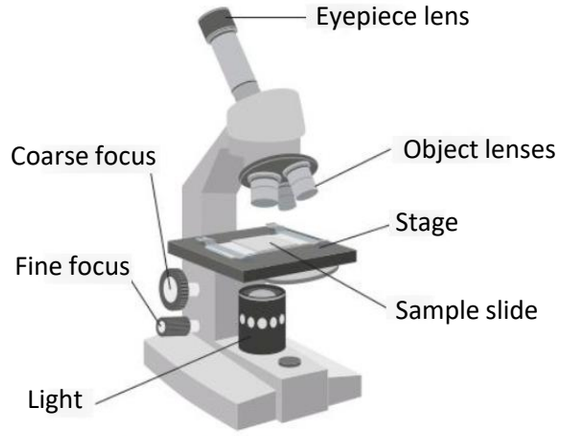
How are plant cells different?
- Plant cells have different cell organelles to an animal cell
- Just like animal cells they have a nucleus, cell membrane, cytoplasm and mitochondria
- In addition they have a **cell wall, vacuole** and **chloroplasts**

Organelle	Function	Found in?
Nucleus	Controls the cell, stores DNA	Animal, Plant
Membrane	Controls what comes in and out	Animal, Plant
Cytoplasm	Site of chemical reactions	Animal, Plant
Mitochondria	Site of respiration	Animal, Plant
Wall	Keeps the cell rigid, protection	Plant
Chloroplast	Site of photosynthesis	Plant
Vacuole	Stores cell sap, keeps the cell firm	Plant

How do we see cells?
- Cells are so small (around 0.01mm long) that we need to use a **microscope** in order to see them
- **Light microscopes** use **lenses** to **magnify** the size of the image, making the sample appear bigger than it really is

Method for using a microscope:
1. Move the stage down to its lowest position
2. Place the slide onto the stage using the clips
3. Select the lowest power objective lens
4. Turn the coarse focus knob slowly until you can see the cells
5. Turn the fine focus knob until the cells are in focus
6. Repeat using the higher power object lens to see the cells in more detail

A labelled diagram of the parts of a microscope:



RAG Yourself!

Learning Outcomes;	RAG
Draw labelled diagrams of an animal cell	
Compare animal and plant cells	
Describe the role of cell organelles	
Label a diagram of a microscope	
Describe how to prepare a microscope slide	
State the levels of organisation in organisms	

Knowledge Organiser
Year 7: Cells – Cell Structure

What are cells?

Levels of organisation within organisms:

How are living things organised?

Level	Examples
Cell	
Tissue	
Organ	
Organ system	
Organism	

What is inside an animal cell?

Labelled diagrams showing the organelles in an animal (left) and plant (right) cell

How are plant cells different?

Organelle	Function	Found in?

How do we see cells?

Method for using a microscope:

1.
2.
3.
4.

5.

6.

A labelled diagram of the parts of a microscope:



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